Current Affairs 10-11-2026

Credit - Drishti Ias

10 Years of Demonetisation in India

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For Prelims: Demonetisation , Reserve Bank of India , Income Tax , Digital Public Infrastructure , Unified Payments Interface , Digital Rupee , Polymer notes

For Mains: Impact of demonetisation on economic growth, liquidity and consumption, Challenges in curbing black money, tax evasion and expanding the direct tax base, Role of the RBI and its powers under the RBI Act, 1934.

Source: TH

Why in News?

The **Demonetisation (2016)** exercise completed 10 years on 8th November 2026. **Reserve Bank of India (RBI)** data highlights its mixed legacy, with significant growth in digital payments, continued **resilience of cash usage and a rise in counterfeiting of new-series currency notes.**

Summary

- 10 Years of Demonetisation accelerated India's digital payment revolution, improved tax compliance and encouraged economic formalisation.
- However, its impact on eliminating black money and reducing cash dependence remains limited, while counterfeiting and disruptions to the informal economy highlight its shortcomings.

What was the Demonetisation of 2016?

- **Demonetisation of 2016:** Demonetisation is the act of **stripping a currency unit of its status as legal tender.**
  - It occurs whenever there is a change of national currency: the current form or forms of money is **pulled from circulation and retired, often to be replaced with new notes or coins.**
  - On 8 th November 2016, the Government of India, acting on the recommendation of the **RBI under Section 26(2) of the RBI Act, 1934** , abruptly withdrew the legal tender status of the existing Rs

500 and Rs 1,000 denomination banknotes.

- **Magnitude:** The withdrawn notes amounted to Rs 15.42 lakh crore, accounting for roughly 86% of all cash in circulation by value at the time.
- **Stated Objectives:** The sudden economic shock was introduced with multiple core aims:
  - Eliminate **"black money" and untaxed, unaccounted wealth.**
  - Curb the circulation of **counterfeit currency** (Fake Indian Currency Notes).
  - Cut off funding for terrorism, naxalism, and subversive activities.
  - Transition the Indian economy towards a **"less-cash" or formalized digital economy.**
- **Operation Clean Money:** Launched by the **Income Tax Department under the Central Board of Direct Taxes (CBDT) in 2017,** aimed to curb **tax evasion** and black money following demonetisation.
  - It used data analytics and e-verification to scrutinise suspicious cash deposits made between **9 th November and 30 th December 2016,** identifying discrepancies between deposited amounts and taxpayers' historical income profiles.
  - The initiative sought to widen the direct tax base, improve tax compliance and promote economic formalisation.
- **Supreme Court Ruling (2023):** In ***Vivek Narayan Sharma v. Union of India (2023)*** , a five-judge Bench upheld demonetisation by a 4:1 majority, finding no legal flaw in the decision-making process and holding that it satisfied the proportionality test.
  - In a dissenting opinion Justice B.V. Nagarathna argued that such a major economic measure should have been **enacted through a plenary legislation (an Act of Parliament) rather than a sudden executive gazette notification.**
- **Withdrawal of Rs 2,000 Notes (2023):** The **RBI withdrew Rs 2,000 banknotes** from circulation under its **Clean Note Policy ,** but they continued to retain legal-tender status, unlike the demonetisation of 2016.

Historical Examples of Demonetisation in India and the World

- **India:**
  - **1946:** India's first demonetisation scrapped Rs 500, Rs 1,000 and Rs 10,000 notes to curb black money accumulated through wartime profiteering during **World War II.**
  - **1978:** The government demonetised Rs 1,000, Rs 5,000 and Rs 10,000 notes under the High Denomination Bank Notes (Demonetisation) Act, 1978, following the **Wanchoo Committee (1970)** recommendations to curb black money.
    - The Committee had proposed demonetising 10, 100, and higher denomination notes, which accounted for 86.6% of the money stock.
- **Global Examples:**
  - **US (1969):** Withdrew high-denomination currency notes above USD 100 to combat organised crime and tax evasion.
    - Unlike conventional demonetisation, existing notes retained their legal tender status, limiting economic disruption.
  - **Soviet Union (1991):** Abruptly withdrew 50 and 100-ruble notes, triggering economic disruption, public distrust and further weakening the government during the Soviet Union's final months.

What have Been the Key Successes and Limitations of Demonetisation?

Key Successes: The Digital Revolution

- **The Digital Revolution:** The most profound structural shift has been the mass adoption of **Digital Public Infrastructure (DPI).**
  - **UPI's Exponential Growth:** From virtually zero in 2016, the value of **Unified Payments Interface (UPI)** transactions has skyrocketed, equating to nearly **91% of India's GDP in 2025-26** , compounded by affordable smartphones, internet connectivity, and changing payment behaviour during Covid-19.
- **Expansion of the Tax Base and Formalisation:** The severe cash squeeze flushed millions of unrecorded transactions into the formal banking system.
  - Initiatives like **Operation Clean Money** helped expand the direct tax base, promote economic formalisation and increase **Income Tax Return (ITR)** filings From just 3.8 crore returns filed in 2013, filings have surged to 8.89 crore for AY 2024-25, marking a 133 percent rise.
- **Short-term Choke on Illicit Networks:** The immediate overnight shock of the note ban successfully paralyzed the financial pipelines of hawala operators, terror-financing networks, and **Left-Wing Extremism (Naxalism) in the short term by turning their illicit cash stockpiles into useless paper.**

Key Limitations and Unmet Objectives

- **The Paradox of Cash (Cash-to-GDP Ratio):** Despite heavy policy pushes toward a digital economy, the public's demand for physical cash has not diminished over the last ten years.
  - In 2015–16, currency with the public stood at 11.9% of GDP, compared with **12.1% in 2025–26** , indicating that cash holdings relative to GDP remained broadly unchanged despite rapid digitalisation.
  - In absolute terms, the total value of banknotes in circulation jumped drastically from around **Rs 16.4 lakh crore in March 2016 to over Rs 41.2 lakh crore by March 2026** .
- **Rise of Counterfeiting in New Notes:** One of the central goals of demonetisation was to wipe out fake notes through the introduction of a highly secure new currency series. However, a decade later, the data reflects a concerning trend:
  - While **demonetisation immediately voided older fakes** , counterfeiting of the new Rs 500 notes which were specifically designed to defeat forgers has skyrocketed.
  - For instance, the RBI reported a **37.3% rise in counterfeit Rs 500 notes detected in 2024–25** , and the total reached **1.42 lakh fake Rs 500s detected in 2025–26** . Within a few years of its launch, the **new Rs 500 became the most counterfeited note in the system.**
- **The Return of Withdrawn Notes:** A core government assumption in 2016 was that a significant portion of black money stored in cash would not be returned to banks due to fear of prosecution, potentially providing the RBI with a windfall dividend.
  - However, over **99% of the invalidated currency** returned to the formal banking system.
  - This proved that black money is predominantly a **"flow" parked in hard assets (real estate, gold, offshore accounts), not just a "stock" of physical cash.**
- **Operational Costs:** RBI expenditure on security printing rose from approximately Rs 3,421 crore in 2015–16 to RS 7,965 crore in 2016–17, alongside additional costs of currency replacement, transportation, ATM recalibration and cash handling.
- **Devastation of the Informal Economy:** The sudden withdrawal of 86% of the country's liquidity **crippled the cash-dependent informal sector.**
  - Agriculture, Micro, Small and Medium Enterprises (MSMEs), and daily-wage laborers faced massive short-term supply chain disruptions, leading to severe **job losses, a temporary contraction in GDP growth and can lead to reverse migration.**

What Measures are Needed for Effective Currency Management?

- **AI-Driven Sorting at Currency Chests:** RBI and commercial banks should deploy Artificial Intelligence and machine-learning-driven sorting machines at regional chests to instantly identify micro-anomalies in fakes and trace their geographical entry points.
- **Scaling the Central Bank Digital Currency (CBDC):** The expansion of the **digital rupee (e₹)** should prioritise offline functionality, interoperability, cybersecurity and privacy safeguards, while ensuring that digital payments complement rather than replace access to physical cash.
  - Programmable **CBDCs** can improve traceability, targeted transfers and payment efficiency.
- **Targeting Asset Sinks via Big Data:** Instead of targeting physical cash, the **Enforcement Directorate (ED)** **and** **Financial Intelligence Unit (FIU)** must integrate databases (GST, PAN, property registrations) via platforms like **Project Insight** to automatically flag disproportionate accumulations in real estate and gold.
- **Safeguarding the Digital Ecosystem:** To prevent MSMEs from reverting to cash, regulators must maintain low **Merchant Discount Rates (MDR)** on digital payments and promote offline digital

solutions (like **UPI-Lite** / **Near Field Communication (NFC)**) for rural financial inclusion.

- **Introduction of Polymer Notes** : The RBI should pilot **Biaxially Oriented Polypropylene (BOPP)-based polymer notes** in lower denominations. Their transparent security features make counterfeiting difficult, while their 2.5–4 times longer lifespan could reduce currency replacement costs under the Clean Note Policy.

Conclusion

The 2016 demonetisation served as a historic catalyst for India's economic formalization. Moving forward, leveraging this robust digital foundation to achieve **SDG 8 (Inclusive Economic Growth)** requires sustained institutional vigilance and seamless technological integration.

Drishti Mains Question:

Ten years after demonetisation, India's economy presents a mixed picture of digital transformation and persistent cash dependence. Critically evaluate its achievements and limitations.

Frequently Asked Questions (FAQs)

1. Under which provision was the 2016 demonetisation exercise carried out?
It was carried out under Section 26(2) of the RBI Act, 1934, following the recommendation of the RBI's Central Board.

2. What was Operation Clean Money?
Launched in 2017 by the Income Tax Department under CBDT, it used data analytics and e-verification to identify suspicious cash deposits following demonetisation.

3. What was the Supreme Court's verdict on demonetisation in 2023?
In Vivek Narayan Sharma v. Union of India (2023), a five-judge Constitution Bench upheld demonetisation by a 4:1 majority. Justice B.V. Nagarathna dissented.

4. What was the major finding regarding returned currency after demonetisation?
Over 99% of the demonetised currency returned to the banking system, challenging the expectation that substantial unaccounted cash would remain unreturned.

5. What is the Central Bank Digital Currency (CBDC)?
CBDC or Digital Rupee (e₹) is the digital form of India's fiat currency, issued by the RBI. It serves as a sovereign digital payment instrument.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/c3VMbFrWwMU**]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Q. Which of the following are the sources of income for the Reserve Bank of India? (2025)

1. Buying and selling Government bonds
2. Buying and selling foreign currency
3. Pension fund management

4. Lending to private companies
5. Printing and distributing currency notes

Select the correct answer using the code given below.

(a) I and II only

(b) II, III and IV

(c) I, III, IV and V

(d) I, II and V

Ans: A

Q. Which of the following measures would result in an increase in the money supply in the economy? (2012)

1. Purchase of government securities from the public by the Central Bank
2. Deposit of currency in commercial banks by the public
3. Borrowing by the government from the Central Bank
4. Sale of government securities to the public by the Central Bank

Select the correct answer using the codes given below:

(a) 1 only

(b) 2 and 4 only

(c) 1 and 3 only

(d) 2, 3 and 4

Ans: (c)

Q. Supply of money remaining the same when there is an increase in demand for money, there will be (2013)

(a) a fall in the level of prices

(b) an increase in the rate of interest

(c) a decrease in the rate of interest

(d) an increase in the level of income and employment

Ans: (b)

Q. A rise in general level of prices may be caused by (2013)

1. an increase in the money supply
2. a decrease in the aggregate level of output
3. an increase in the effective demand

Select the correct answer using the codes given below:

(a) 1 only

(b) 1 and 2 only

(c) 2 and 3 only

(d) 1, 2 and 3

Ans: (d)

Bangladesh's RMG Crisis: A Strategic Opportunity for India

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For Prelims: Readymade Garment , Least Developed Country , CEPA , ECTA, PM MITRA , SAMARTH scheme

For Mains: Textile Sector of India, Potential and Challenges, Growth & Development

Source: IE

Why in News?

Massive protests by **Ready-Made Garment (RMG)** workers across Bangladesh's industrial belts, particularly Gazipur, Ashulia and Dhaka, have forced nearly 700 factories to temporarily shut down. Occurring during the critical peak-order season for Western markets, this labour unrest alongside shifting global trade dynamics, presents a **critical strategic opening for India's textile and Readymade Garment (RMG) sector.**

Summary

- Bangladesh's RMG crisis, driven by labour unrest, rising costs and factory closures, creates an opportunity for India to attract global apparel orders and expand its export market share.
- India must strengthen textile infrastructure through PM MITRA, promote MMF manufacturing through PLI, improve logistics, rationalise duties and enhance labour skills to become a competitive global textile hub.

What is the Current Crisis in Bangladesh's Textile Sector?

Rise of Bangladesh's RMG Sector:

- **Historical Shift:** Initially an agrarian economy reliant on the jute trade in 1971, Bangladesh strategically pivoted **toward garment manufacturing to build its industrial base.**
- **Advantage of LDC Status:** Under the **Multi-Fibre Arrangement (1974-1994)** (an international

trade framework that governed the global trade in textiles and clothing) **,** Western nations imposed import quotas on booming producers like South Korea and Hong Kong.

  - Bangladesh leveraged its **Least Developed Country (LDC) exemption** . This status granted **Duty-Free and Quota-Free (DFQF) access** to lucrative Western markets **such as the EU's "Everything But Arms" scheme,** aggressively redirecting foreign capital and textile manufacturing to Dhaka.

- **Unmatchably Cheap Labour:** This manufacturing influx was sustained by an **"inventory-per-day" model** heavily reliant on one of the lowest wage structures globally, ultimately **propelling Bangladesh to become the world's second-largest apparel exporter behind China.**
- **Economic Backbone:** As of 2026, the RMG sector employs over 4 million workers, generates approximately USD 40 billion annually, and **accounts for roughly 80% of Bangladesh’s total export revenue.**

Causes of Current RMG Crisis in Bangladesh

- **Wage–Cost of Living Gap:** Workers demand a wage hike as inflation erodes purchasing power and current wages fall short of living costs.
- **Energy Crisis:** Declining domestic gas production and expensive **LNG** imports have pushed RMG production costs up by nearly 40% in three years.
- **Factory Closures:** Over 400 garment factories shut down (July 2023–June 2026) due to financial stress, energy shortages and declining orders, causing massive job losses.

How Does the Bangladesh RMG Crisis Present an Opportunity for India?

- **Supply Chain De-risking (The 'Bangladesh Plus One' Strategy):** Global fashion conglomerates lock in 40-50% of their annual inventory between November and February. The unrest creates a severe supply-chain shock.
  - Bangladesh's annual RMG exports of USD 40–45 billion present an opportunity for India. Capturing 10–15% of diverted trade could boost **India's apparel exports by USD 4–5 billion annually and raise its global RMG market share beyond the current 3–4%.**
- **The 'LDC Graduation' Tariff Arbitrage:** Bangladesh is slated to graduate from its **UN LDC status by 2026 (with a grace period up to 2029).**
  - Consequently, it will lose its DFQF access under the EU’s "Everything But Arms" (EBA) scheme.
  - Post-graduation, Bangladeshi garments will face standard import tariffs of **9.6% to 12%** in European markets.
  - **India's FTAs with the UK and EU** , along with **CEPA with the UAE** and **ECTA with Australia** , strengthen its export competitiveness and reduce Bangladesh's traditional tariff advantage.
- **Complete 'Farm to Fashion' Value Chain:** Bangladesh imports nearly 90% of its cotton, leaving it highly vulnerable to forex depletion and global price shocks.
  - In contrast, India is among the **largest producers of cotton and silk** , possessing a **fully integrated, indigenous supply chain from spinning and weaving to garmenting.**
- **Leveraging Established Textile Clusters:** Major textile centres such as Tiruppur (Tamil Nadu), Surat (Gujarat) and Gautam Buddha Nagar (Uttar Pradesh) possess established manufacturing ecosystems that can help absorb additional orders from global apparel buyers.
  - Tiruppur's knitwear cluster is particularly well-positioned to benefit from international apparel sourcing

diversification.

- **Policy and Infrastructure Impetus:** India is aggressively mitigating its historical infrastructure deficit through the **PM MITRA (Mega Integrated Textile Region and Apparel)** scheme to create plug-and-play infrastructure.
  - The **PLI (Production Linked Incentive) Scheme for Textiles** is specifically targeted at bridging **India's gap in Man-Made Fibres (MMF) and technical textiles, aligning with global demand.**
- **Macroeconomic and ESG Stability:** Amidst Bangladesh's plunging forex reserves (hovering below USD 20 billion in usable reserves) and energy rationing, India offers a highly stable currency (INR), robust **forex reserves** (over USD 680 billion) stable environment for long-term capital expenditure by global brands.

Current Status and Strategic Importance of India's Textile Sector

- **Macroeconomic Contribution:** As of February 2026, The Indian textile and allied sectors contribute approximately **2% to GDP, 11% to manufacturing GVA and 8.63% to total exports.**
- **Employment Generation:** It is a critical livelihood provider, directly employing over **45 million (4.5 crore) workers** , heavily supporting women and rural populations.
- **Global Trade Position:** In 2025, India is the **6 th largest exporter of textiles and apparel globally** , holding a **3.9% share in global trade** , with key export destinations being the US, EU, and UAE.
- **Raw Material Sovereignty:** India is the world's largest cotton cultivator by acreage, the **largest jute producer** , and the **2nd-largest producer of silk, polyester, viscose fibres and Man-Made Fibres (MMF), after China.**
- **Investment Climate:** To boost global integration, **100% Foreign Direct Investment (FDI) is allowed under the automatic route** .



What are the Associated Structural Challenges for India?

- **Competition from Established Export Hubs:** Countries such as Vietnam, Cambodia and Indonesia compete with India through established apparel-manufacturing ecosystems, international buyer networks and preferential trade arrangements.
  - Vietnam (holding a ~6% global market share) is currently the primary beneficiary of the **'China Plus One'** and **'Bangladesh Plus One' shifts.**
  - Vietnam boasts a highly integrated supply chain ecosystem with a lead time of just **15–21 days** , compared to India’s fragmented supply chain which often takes **45–60 days** from order to shipping.
  - Furthermore, Vietnam already enjoys zero-duty access to the EU under the **EVFTA (EU-Vietnam Free Trade Agreement)** .
- **The "Missing Middle":** Unlike the massive, consolidated 2,000+ worker mega-factories in Bangladesh and Vietnam. India's textile sector is dominated by Micro, Small and Medium Enterprises (MSMEs), limiting economies of scale, production standardisation and the capacity to fulfil large international orders.
  - Establishing and scaling new garment-manufacturing facilities could **take 3–5 years, restricting India's ability to absorb diverted orders immediately.**
- **The Man-Made Fibre (MMF) Disconnect:** Global fast fashion is heavily **synthetic-driven (70% MMFto 30% cotton**). Conversely, India’s export basket is inverted, dominated by cotton (**70% cotton to 30% MMF**).
  - Furthermore, **inverted duty structures and anti-dumping duties on synthetic raw materials**

make Indian MMF apparel uncompetitive in global pricing.

- **Boomerang Effect on Domestic Spinning Mills:** Because South Asian supply chains are deeply interdependent, the crisis hurts India directly.
  - Bangladesh absorbs roughly 45% of India's total cotton yarn exports (over USD 1.5 billion), meaning **Dhaka's factory closures translate to immediate order cancellations and massive inventory pile-ups for Indian mills.**
- **Technology and Productivity Gaps:** Limited adoption of automated cutting, computer-aided design (CAD), enterprise resource planning (ERP) and digital production systems among smaller manufacturers constrains productivity, quality consistency and delivery efficiency.

What Measures Can Strengthen the India RMG Sector and Become a Global Hub?

- **Fast-Track PM MITRA Parks:** Rapidly operationalize integrated mega textile parks to cluster the entire value chain. This will slash logistics costs to the 8% global standard and drastically improve **Turnaround Time (TAT).**
- **Enhance ESG Compliance:** Proactively market India's democratic labor laws and green technologies (e.g., zero-liquid discharge) to meet strict **Western sustainability mandates, such as the EU's Carbon Border Adjustment Mechanism (CBAM).**
- **Boost Female Employment via 'SAMARTH':** Leverage the **SAMARTH scheme** to upskill rural women, replicating the demographic dividend of Bangladesh's RMG boom while improving India's **Female Labor Force Participation (FLFP).**
- **Rationalize Duty Structures:** Correct the inverted GST and import duties on Man-Made Fibre (MMF) raw materials to achieve global price parity and capture the lucrative fast-fashion market.
- **Promote Technology-Led Productivity** : Encourage adoption of **(Computer-Aided Design / Computer-Aided Manufacturing) and Industry 4.0 technologies** through targeted technology-upgradation support to improve manufacturing productivity, quality and scalability

Conclusion

India, currently the world's sixth-largest apparel exporter, relying solely on a neighbor's crisis is not a sustainable long-term strategy. By scaling up capacities through PM MITRA, prioritizing Man-Made Fibres, and fully utilizing newly negotiated FTAs, India can transform this temporary geopolitical opening into a permanent position as the globe's premier textile manufacturing hub.

Drishti Mains Question:

Global supply chain diversification presents an opportunity for India to emerge as an alternative manufacturing destination. Discuss in the context of the China Plus One and Bangladesh Plus One strategies.

Frequently Asked Questions (FAQs)

1. What is the significance of the Ready-Made Garment (RMG) sector in Bangladesh's economy?
The RMG sector employs over 4 million workers, generates approximately USD 40 billion annually and contributes nearly 80% of Bangladesh's export revenue.

2. What is the significance of LDC graduation for Bangladesh's textile exports?
Graduation from Least Developed Country (LDC) status will lead to the eventual loss of preferential market access, including duty-free access under the EU's Everything But Arms (EBA) scheme, subject to transitional arrangements.

3. What is the objective of the PM MITRA Scheme?
The PM Mega Integrated Textile Regions and Apparel (PM MITRA) Scheme aims to establish integrated textile parks with world-class infrastructure and promote economies of scale across the textile value chain.

4. What is the objective of the PLI Scheme for Textiles?
The Production Linked Incentive (PLI) Scheme for Textiles promotes domestic manufacturing of Man-Made Fibre (MMF) apparel, MMF fabrics and technical textiles to enhance global competitiveness.

5. What is the SAMARTH Scheme?
The Scheme for Capacity Building in Textile Sector (SAMARTH) aims to provide skill development and placement-oriented training to workers in the textile and apparel industry.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/P4l8mpALVss**]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. Consider the following statements: (2020)

1. The value of Indo-Sri Lanka trade has consistently increased in the last decade.
2. “Textile and textile articles” constitute an important item of trade between India and Bangladesh.
3. In the last five years, Nepal has been the largest trading partner of India in South Asia.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 only
(c) 3 only
(d) 1, 2 and 3

Ans: (b)

Mains

Q. Analyse the factors for highly decentralised cotton textile industry in India. **(2013)**

Physical AI and its Applications

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Source: IE

Why in News?

Major tech and energy firms, including OpenAI and ADNOC (Abu Dhabi National Oil Company), signed an agreement at the **Abu Dhabi International Petroleum Exhibition and Conference (ADIPEC) 2026 to leverage artificial intelligence (AI) for operational efficiency in the oil sector.**

- This marks a definitive shift toward **"Physical AI,"** a trend Nvidia framed as the **"ChatGPT moment for robotics,"** aiming to address the AI sector's core challenge of monetisation by deploying AI in real-world industrial systems.

What is Physical AI?

- **About:** Physical AI refers to AI systems designed to operate in and interact with the physical world (the realm of "atoms") rather than existing purely in digital environments (like chatbots or code assistants).
  - It enables autonomous systems such as **robots, camera-bots, and self-driving cars to perceive their surroundings** , reason about them, and perform complex physical actions.
- **Mechanism:** Physical AI integrates **large language models (LLMs)** , sensors (cameras, **LiDAR**), control systems and actuators to perceive, process, plan and execute physical actions.
  - It uses computer vision, reinforcement learning, imitation learning and simulation-based training to adapt to real-world environments.
- **Evolution from Traditional Robotics:** While traditional factory robots follow rigid, pre-programmed paths and fail if an object shifts even slightly.
  - Physical AI enhances autonomy, real-time adaptability and decision-making in dynamic environments. For example, AI-powered robotic arms automatically adjust their gripping force and position based on an object's shape and orientation on a conveyor belt.
- **Significance of Physical AI:**
  - **Solving the Monetisation Bugbear:** By transitioning into the B2B enterprise and industrial sectors, physical AI provides concrete Return on Investment (RoI) for tech companies struggling to monetise consumer-facing digital AI tools.
    - Analysts at McKinsey project that robotics and physical AI will create at least a **trillion dollars in economic value by 2040,** primarily driven by efficiencies in manufacturing and logistics.
  - **Enhanced Productivity:** It shifts manufacturing from rigid automation to flexible production lines.
    - For example, AI-assisted production scheduling systems have helped manufacturers **reduce non-value-added time by up to 75%.**
  - **Workplace Safety and Ergonomics:** Physical AI handles dangerous tasks, while innovations like **robotic exoskeletons support assembly-line workers during physically demanding overhead tasks** , significantly reducing fatigue and injury.

- **Continuous Learning Architecture:** AI tools deployed in real-world systems log joint angles and sensor feeds, allowing the **models to progressively learn and get better with continuous use.**
- **Simulation-Based Learning and Cost Reduction:** Physical AI employs digital twins (virtual replicas of physical systems) and synthetic data to train and test autonomous machines in simulated environments, reducing dependence on expensive and potentially hazardous real-world training.

- **Concerns:**
  - **Capital Intensity and Hardware Costs:** Developing and deploying high-fidelity robotic arms, tactile sensors, and mobile platforms remains highly capital-intensive.
  - **Safety and Regulation:** Integrating autonomous, learning machines into human workspaces raises immense safety concerns.
    - Emerging frameworks, such as the **EU Machinery Regulation 2023 (applicable from 2027) and the EU AI Act 2024,** are mandating strict safety and compliance standards for **AI in physical workspaces.**
  - **Societal Acceptance:** While widely seen as transformational, scaling physical AI (especially humanoid robots) faces hurdles regarding societal acceptance and the persistent **fear of blue-collar job displacement.**
  - **Simulation-to-Reality Gap:** AI models trained in simulated environments may perform unpredictably under unfamiliar real-world conditions, while inadequate training data can leave rare or safety-critical situations unaddressed.

Key Industrial Applications:

- **Subsurface Energy Exploration: Chevron** (US-based multinational energy company) launched its in-house generative AI tool, **ApEX,** which uses **geospatial agents to interpret seismic data** and assess prospective drill sites in the **Gulf of Mexico.**
  - **British Petroleum (UK-based energy company)** uses AI to analyse real-time drilling data, reducing exploration costs by 7% and non-productive time by nearly 8%.
- **Predictive Maintenance (Shell):** Shell (UK-based multinational oil and gas company) implemented an enterprise-scale **predictive maintenance program** using machine learning (C3.ai) to monitor over 13,000 pieces of equipment, effectively preventing failures and reducing emergency repair costs.
- **Advanced Manufacturing: Foxconn** (Taiwan-based electronics manufacturing company) is utilizing a "scalable AI-powered robotic workforce" paired with digital twin simulations.
  - This enables robots to handle precise, variable tasks like screw tightening and cable insertion, reducing system deployment times by nearly 50% and lowering error rates by 25%.
- **Logistics and Ports:** AI systems are increasingly managing infrastructure, such as **India's Vizhinjam port,** where AI assists in monitoring ship movements, flagging navigational hazards, and coordinating automated cranes.

Frequently Asked Questions (FAQs)

1. What is Physical AI?
Physical AI refers to AI systems that perceive, reason and interact with the physical world through robots, autonomous machines and real-time sensor data.

2. How does Physical AI differ from traditional robotics?
Traditional robots follow pre-programmed instructions, whereas Physical AI-powered robots can learn, adapt to changing environments and make real-time decisions.

3. What is the role of Digital Twins in Physical AI?
Digital Twins create virtual replicas of physical systems to simulate, test and optimise industrial operations before real-world deployment.

4. Which major companies are adopting Physical AI in industrial operations?
Chevron (USA), BP (UK), Shell (UK) and Foxconn (Taiwan) are deploying AI in exploration, predictive maintenance and advanced manufacturing.

[Watch Video on YouTube: ▶ https://www.youtube.com/embed/hfXyDIsRdCQ]

UPSC Civil Services Examination Previous Year Question (PYQ)

Prelims

Q. With the present state of development, Artificial Intelligence can effectively do which of the following? (2020)

1. Bring down electricity consumption in industrial units
2. Create meaningful short stories and songs
3. Disease diagnosis
4. Text-to-Speech Conversion
5. Wireless transmission of electrical energy

Select the correct answer using the code given below:

(a) 1, 2, 3 and 5 only

(b) 1, 3 and 4 only

(c) 2, 4 and 5 only

(d) 1, 2, 3, 4 and 5

Ans: (b)

Mains

Q. Introduce the concept of Artificial Intelligence (AI). How does AI help clinical diagnosis? Do you perceive any threat to privacy of the individual in the use of AI in healthcare? **(2023)**

Q. How globalization has led to the reduction of employment in the formal sector of the Indian economy? Is increased informalization detrimental to the development of the country? **(2016)**

SC Rules no Distinction Between Arrest and Detention

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Source:IE

In the landmark ***Sujit Das and Others v. Union of India*** **(November 2026)** case, the Supreme Court ruled that there is no constitutional distinction between **"arrest" and "detention,"** declaring that the **mandatory 24-hour period for judicial production under Article 22(2) begins the exact moment a person's liberty is curtailed** , rather than when the police formally record the arrest.

- **Conjunctive Reading of Article 22(2):** The Court clarified that the phrase **“arrested and detained in custody” is conjunctive and synonymous** . Authorities cannot create an artificial distinction between the two terms to delay presenting the accused before a Magistrate.
- **Elimination of "Informal Custody":** The bench ruled that holding individuals for **"prolonged interrogation" or "search and seizure" without recording a formal arrest is unconstitutional** .
  - There is absolutely no legal "buffer period" preceding an arrest.
- **Resetting the 24-Hour Clock:** The constitutional **24-hour countdown starts immediately upon actual detention** .
  - The only permissible exclusion from this time frame is the journey time to the Magistrate's court.
  - **Article 22(2) of the Indian Constitution provides** that every arrested person must be produced before the **nearest magistrate within 24 hours of arrest, excluding travel time.** Further detention requires the magistrate’s authorisation.
    - This safeguard does not apply to enemy aliens and persons detained under preventive detention laws, as **provided under Article 22(3).**
- **Immediate Release upon Violation:** If a suspect is held beyond 24 hours without Magisterial production, it is a direct **violation of fundamental rights , entitling the person to be released forthwith** .
- **Reasonableness of Detention (Section 58, BNSS):** Relying on **Section 58 of the Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023** (formerly Sec 57 CrPC), the Court noted that simply adhering to the 24-hour limit does not make a detention automatically lawful.
  - The **Magistrate must actively verify if the detention within those 24 hours was "reasonable."** Magistrates can decline police remand if there is an unexplained, unreasonable gap (e.g., 18 hours in informal custody) between physical detention and formal arrest.
- **Primacy of Fundamental Rights:** Statutory restrictions on bail under special laws (like the **NDPS Act, 1985**) **cannot override the fundamental rights guaranteed under Article 21 (Right to Life and Liberty) and Article 22** . The Constitution dictates the interpretation of the statute, not vice versa.

- **CCTV Surveillance:** Reiterating **Paramvir Singh Saini v. Baljit Singh (2020),** the Court emphasised audio-video CCTV surveillance with night vision in police stations and offices of agencies such as the **CBI , ED , and NIA** where interrogation and detention take place.
- **Significance & Impact:** This ruling acts as a major safeguard against **custodial violence and torture** , strictly **limits arbitrary state action** , and ends the mechanical granting of remands by **shifting the burden of active oversight onto Magistrates** .

Read more: Preventive Detention

- **SC Mandates Strict Safeguards for Arrest and Re-arrest**

3 Indian Manuscripts Added to UNESCO's Memory of the World Asia-Pacific Register

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Source: AIR

Recently, **three major works** of India's literary and intellectual heritage— **Kalhana's Rajatarangini, the Mahabharata and Kautilya's Arthashastra** —were added to **UNESCO's Memory of the World Regional Register for Asia and the Pacific.**

- **Recognition:** The decision was taken at the **11th General Meeting of the Memory of the World Committee for Asia and the Pacific (MOWCAP)** , held in **Jakarta, Indonesia** .
- **India's Total Inscriptions:** With these additions, India's total number of inscriptions in the regional register has **increased to six.**
  - The previous three, inscribed in 2024, were **Ramcharitmanas, Panchatantra and Sahrdayaloka-Locana.**
- **Rajatarangini:** Written by **Kalhana in the 12 th century,** it is a Sanskrit historical chronicle of Kashmir, documenting its rulers and political history. The manuscript is preserved at the **Oriental Research Library, Srinagar.**
- **Mahabharata:** Traditionally attributed to **Vyasa, it is one of India's two major Sanskrit epics,** exploring Dharma, duty and morality. The inscribed **400-year-old birch-bark manuscript,** written in **Sharada script,** is preserved at the **Bhandarkar Oriental Research Institute, Pune.**
- **Arthashastra** : Traditionally attributed to **Kautilya (Chanakya),** it is an ancient treatise on governance, administration, diplomacy, economics and statecraft. Its inscribed 16th-century palm-leaf manuscript is preserved at the **Oriental Research Institute, Mysuru.**
- **Role of IGNCA:** The **Indira Gandhi National Centre for the Arts (IGNCA)** played an important role in preparing the nomination dossiers for all three manuscripts .

UNESCO Memory of the World Programme

- **Established and Objective:** Launched by **UNESCO** in 1992, the programme aims to preserve documentary heritage, particularly that threatened by **conflicts and natural disasters, ensure universal access and enhance public awareness of its significance** .
- **Registers:** The programme operates through **international, regional and national registers,** covering documentary heritage such as manuscripts, archives and audiovisual records.
- **UNESCO/Jikji Memory of the World Prize:** Established in **2004,** it recognises outstanding contributions to the preservation and accessibility of documentary heritage.
  - It is named after **Buljo Jikji Simche Yojeol (Jikji),** the world's **oldest surviving book** printed using movable metal type.

Read more: **UNESCO's Memory of the World Asia-Pacific Regional Register**

FATF Steps Up Action Against Evolving Financial Crime Threats

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Source: AIR

Recently, the **Financial Action Task Force (FATF)** Plenary meeting in Paris, under the UK Presidency, agreed to intensify global action against emerging financial crime threats, particularly the **misuse of virtual assets, financial fraud and scam compounds.**

- **Virtual Assets Regulation:** Following the **FATF Ministerial Declaration of April 2026,** the Plenary agreed to launch a targeted review and accelerated follow-up process to strengthen implementation of FATF standards on **virtual assets (VAs)** **and Virtual Asset Service Providers (VASPs)** in jurisdictions with significant virtual asset activity.
  - The review findings are expected in **June and October 2027,** focusing on risks related to money laundering, terrorist financing and proliferation financing.
- **Combating Fraud and Scam Compounds:** FATF approved a report examining illicit financial flows linked to scam compounds, **aiming to disrupt organised criminal networks and the emerging crime-as-a-service economy** under its Roadmap on **Combating Fraud.**
- **Removal from Increased Monitoring: Bulgaria, Côte d'Ivoire, the Democratic Republic of the Congo and Monaco** were removed from **FATF's Grey List** after completing their action plans and undergoing successful on-site assessments.
  - FATF also expressed concern over **Vietnam's slow progress in addressing strategic deficiencies** in its anti-money laundering and counter-terrorist financing framework.

About FATF

- **Establishment:** FATF is an intergovernmental organisation **established in 1989** at the initiative of the **G7 countries** , with headquarters in Paris, France.
- **Mandate** : It is the global watchdog against **money laundering** , terrorist financing and proliferation financing. Initially focused on money laundering, its mandate expanded to include terrorist financing in 2001 and proliferation financing in 2012.
- **Decision-Making Body:** The **FATF Plenary** is its decision-making body and generally meets three times a year.
- **Membership** : FATF has 40 members, comprising 38 jurisdictions and two regional organisations, the **European Commission** and **Gulf Cooperation Council** (GCC). Russia's membership has remained suspended since February 2023.
- **India and FATF:** India became an observer in 2006 and a full member in 2010. It is also a member of the **Asia/Pacific Group (APG) and Eurasian Group (EAG)** , both FATF-style regional bodies.
  - Vivek Aggarwal of India assumed the FATF Vice-Presidency on 1 st July 2026.
- **FATF Monitoring Lists:**
  - **Grey List:** Jurisdictions under increased monitoring that have **committed to addressing strategic deficiencies** in combating money laundering, terrorist financing and proliferation financing within agreed timelines.
  - **Black List:** High-risk jurisdictions subject to a call for action due to **serious deficiencies in combating money laundering** , terrorist financing and proliferation financing.
    - As of October 2026, it includes **North Korea (DPRK), Iran and Myanmar.**

Read more: **Financial Action Task Force (FATF)**

Exercise Naseem-Al-Bahr

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Source: DD

Recently, **Indian Naval Ship (INS) Surat** arrived at Muscat, Oman, to participate in the **15 th edition** of the bilateral maritime exercise **Naseem Al Bahr (Sea Breeze)** , followed by a **Coordinated Patrol (CORPAT)** between the **Indian Navy and the Royal Navy of Oman.**

- **About:** Initiated in **1993, Naseem Al Bahr** is a biennial bilateral naval exercise between **India and Oman** aimed at **strengthening maritime cooperation and operational interoperability.**
- **Harbour and Sea Phases:** The exercise includes cross-deck visits, operational interactions and sporting activities, facilitating professional exchanges, sharing of best practices and enhanced interoperability between the two navies.
- **Coordinated Patrol (CORPAT):** Following the exercise, both navies will undertake a coordinated patrol to strengthen maritime surveillance and operational coordination.
- **Significance:** The exercise strengthens India–Oman maritime cooperation, builds on the 13th Joint Military Cooperation Committee (JMCC) meeting held in November 2025, and aligns with **India's MAHASAGAR (Mutual and Holistic Advancement for Security and Growth Across Regions) vision.**
- **INS Surat:** It is the fourth and final **Project 15B (Visakhapatnam-class)** stealth guided-missile destroyer of the Indian Navy, commissioned in 2025..
  - It is equipped with advanced weapons, sensors and surveillance systems, enhancing India's maritime defence capabilities.

Read more: **Exercise Naseem Al Bahr**

Exercise RAHAT MITRA

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Source: DD

Recently, **India and Myanmar** concluded their inaugural **Tri-Services Humanitarian Assistance and Disaster Relief (HADR) exercise, RAHAT MITRA** , in Chennai.

- **Theme and Participation:** The exercise, themed **“Together for Humanity, Stronger through Partnership”** , involved over **300 personnel** from the defence forces of both countries and was conducted under the aegis of the **Indian Navy** .
- **Joint Disaster Response Planning:** A **Table-Top Exercise (TTX)** based on a severe cyclone scenario focused on joint planning, coordinated disaster-response mechanisms and operational interoperability.
- **Multi-Domain HADR Operations:** The exercise showcased land, maritime and air-based disaster-response capabilities, including flood relief, **maritime Search and Rescue (SAR)** , casualty evacuation and medical triage, through demonstrations involving **INS Sukanya, INS Bitra, Chetak helicopter and Advanced Light Helicopter (ALH).**
- **Significance:** The exercise enhances India–Myanmar defence ties, improves joint disaster-response capabilities and reinforces **India's Neighbourhood First policy** and **MAHASAGAR vision** for regional security and cooperation.

Read more: **India-Myanmar Ties: Fencing The Free Movement**